

22. (3 points)

An insurance company needs to strengthen its capital position in order to maintain its financial strength rating. The company currently has no reinsurance and is considering the following two strategies:

Strategy 1: Issue an \$80 million surplus note

- Coupon rate on surplus note = 8%.
- Bond yield = 5%.
- Coupons are paid out annually.
- The surplus note cannot be repaid for a minimum of 5 years.

Strategy 2: Purchase quota-share reinsurance

- Gross earned premium = \$200 million.
- Economic margin on contract = 15% of ceded premium.
- The insurance company will cede 10% of its earned premium in the first year.
- It is projected that the company's profits will strengthen its surplus over time and the cession will be reduced to 8% in the second year, 3% in the third year, and eliminated in the fourth year.
- The projected annual gross earned premium is stable over the next five years.

a. (0.75 point)

Calculate the cost of each strategy in the first year and determine which strategy is preferred on a one-year basis.

b. (1.25 points)

Calculate the cost of each strategy over five years, using the bond yield as a discount rate, and determine which strategy is preferred on a five-year basis.

c. (1 point)

Identify the three paradigms for measuring the value in a reinsurance structure and identify which of the paradigms was used to evaluate the quota share reinsurance structure in Strategy 2 above.

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SPRING 2018 EXAM 7, QUESTION 22

TOTAL POINT VALUE: 3

LEARNING OBJECTIVE(S): C1, C2

SAMPLE ANSWERS

Part a: 0.75 point

Sample 1

Strategy 1: $80 \times (0.08 - 0.05) = 2.4$ million in 1st year

Strategy 2: $200 \times 0.1 \times 0.15 = 3$ million in 1st year

On a 1-year basis, strategy 1 is preferred.

Sample 2

St1.

Coupon Payout $80 \times 0.08 = 6.4$

Bond yield = $80 \times 0.05 = 4$

Cost = $6.4 - 4 = 2.4$ M

St2.

Cost = $200\text{M} \times 10\% \times 15\% = 3\text{M}$

Strategy 1 is preferred.

Sample 3

Cost of surplus note in 1st year:

$80 \text{ million} \times (8\% - 5\%) / 1.05 = 2.286 \text{ million}$

Cost of reinsurance in 1st year:

$200 \text{ million} \times 10\% \times 15\% / 1.05 = 2.857 \text{ million} > 2.286 \text{ million}$

Surplus note is preferred on one-year basis.

Part b: 1.25 points

Sample 1

Option 1

$$\frac{2.4}{1.05} + \frac{2.4}{1.05^2} + \frac{2.4}{1.05^3} + \frac{2.4}{1.05^4} + \frac{2.4}{1.05^5} = 10.39$$

Option 2

$$\frac{3M}{1.05} + \frac{2.4}{1.05^2} + \frac{0.9}{1.05^3} = 5.81$$

Option 2 is preferred on a 5-year basis.

Sample 2

	Yr1	2	3	4	5	Total
#1 Cost	2.4	2.4	2.4	2.4	2.4	
#1 Discount	$2.4/1.05=2.29$	$2.4/1.05^2=2.18$	2.07	1.97	1.88	10.39

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	Yr1	2	3	4	5	Total
#2 Cost	3M	$200 \times 15\% \times 8\% = 2.4$	0.9	0	0	
#2 Discount	$3/1.05 = 2.86$	$2.4/1.05^2 = 2.18$	0.78	0	0	5.81

5.81 < 10.39, so #2 strategy is preferred.

Sample 3

Cost of surplus note over 5 years:

$$80 \times (8\% - 5\%) / 1.05 + \dots + 80 \times (8\% - 5\%) / 1.05^5 = 10.39 \text{ million}$$

Cost of reinsurance over 5 years:

$$200 \times 10\% \times 15\% / 1.05 + 200 \times 8\% \times 15\% / 1.05^2 + 200 \times 3\% \times 15\% / 1.05^3 = 5.811 \text{ million}$$

The reinsurance is preferred on a 5-year basis.

Sample 4

Strategy 1

$$\text{Cost} = 80M - (0.08)(80M)/1.05 + (0.08)(80M)/1.05^2 + \dots + 0.08(80M)/1.05^5 - (80)/1.05^5 = -10.4M$$

Strategy 2

$$0.15(0.10)(200) + 0.15(0.08)(200)/1.05 + 0.15(0.08)(200) / 1.05^2 = 6.1M$$

Strategy 2 preferred.

Part c: 1 point

Sample 1

- Reinsurance provides stability
 - Reinsurance frees up capital
 - Reinsurance adds to firm equity value
- Second paradigm: Frees up capital was used

Sample 2

Is stable

Surplus release

Market value added

Sample 3

Three paradigms are stability, frees up capital, and adding values.

Frees up capital by using less cost to cover the possible exposures

Sample 4

-Stability – of earnings, capital, etc.

-Decrease in required capital

-Adds value to the firm

This strategy will result in a decrease in required capital

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Sample 5

- Stabilize results
- As a replacement for required surplus
- Adds value to the firm
- As a replacement for required surplus

Sample 6

- Reinsurance adds stability to results
 - Reinsurance creates firm value
 - Reinsurance helps manage surplus
- Paradigm 3 was used to evaluate the structure of Strategy 2

EXAMINER'S REPORT

Candidates were expected to demonstrate general knowledge of how to assess the costs associated with strategies used to mitigate insurance risk. They were expected to demonstrate awareness of common Enterprise Risk Management concepts underlying capital management.

Part a

Candidates were expected to know how to calculate the 1st year cost of issuing a surplus note and the cost of purchasing reinsurance, and to be able to identify which strategy would be preferred based on the cost of each.

Common mistakes included:

- Calculating only the value of the coupon payment as the associated surplus note cost for strategy 1.
- Using the economic margin on the reinsurance contract incorrectly to calculate the cost of strategy 2.

Part b

Candidates were expected to know how to calculate the cost of each of the two strategies on a 5-year basis, and be able to identify that the lower cost option would be preferred. Candidates were expected to be able to discount the cost of each strategy using the bond yield that was provided.

Common mistakes included:

- Not discounting the cost of each strategy.
- Using the coupon rate as the discount rate.
- Using the surplus note principal repayment as part of the cost of employing strategy 1.

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Part c

Candidates were expected to be able to identify the three paradigms commonly used to assess the economic value of a reinsurance contract and to be able to identify which of the three was used to evaluate the quota share reinsurance strategy

Common mistakes included:

- Providing two items that were both related to the same paradigm.
- Incorrectly identifying which paradigm was used to evaluate the quota share reinsurance strategy.